

**Econ 630, Spring 2011:
Final Exam**

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Name: _____

The exam consists of 4 questions. Your answer should be clearly labeled and legible. Show all work. No credit will be given for answers that are not justified.

1. (20 pts) Consider the following simultaneous move game:

		Pl 2			
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Pl 1	<i>A</i>	1, 1	2, 0	0, 2	-1, 3
	<i>B</i>	3, -1	1, 1	1, 1	2, 0
	<i>C</i>	2, 0	1, 1	-1, 3	0, 2
	<i>D</i>	0, 2	1, 1	-2, 4	1, 1

Figure 1

- (a) (5 pts) What pure strategies survive iterated strict dominance? Carefully explain each round of elimination (i.e. which strategy you are eliminating as well as the strategy that dominates it).

(b) (5 pts) What is the set of rationalizable pure strategies for player 1 and 2. Carefully justify your answer.

(c) (10 pts) Find all Nash equilibria (pure and mixed).

2. (30 pts) Consider a negotiation game of complete and perfect information between a land developer (b) and two sellers (s_1, s_2) of adjacent land parcels. The buyer has joint valuation for both parcels of $V = 1$ and stand-alone valuation of $v_1 = v_2 = 0$ (i.e. the two parcels are perfect complements). Each seller values his land at $v_s \in (0, \frac{1}{3})$ (the same for both sellers). The game proceeds as follows:

- I The buyer approaches s_1 . The bargaining lasts only one period. Each party is equally likely to make a take-it-or-leave-it offer. With probability $1/2$ the buyer makes an offer $p_1(b)$, which s_1 chooses whether to accept or reject. Alternatively, with probability $1/2$ s_1 makes a take-it-or-leave-it offer $p_1(s_1)$, which the buyer chooses whether to accept or reject. Let $\psi_1 = \{1, 0\}$ denote the outcome of the first negotiation, where 1 stands for the event, in which the buyer has purchased the first parcel, and 0 stands for the event, in which the buyer has not purchased the first parcel. ψ_1 is observable to all players in the game.
- II The buyer chooses whether to approach s_2 . Upon approaching s_2 , the bargaining is identical to the first one: each party makes a take-it-or-leave-it offer with probability $1/2$.

The game ends after the second negotiation. The players are perfectly patient and do not discount the future payoffs (i.e. $\delta = 1$). Moreover, assume that visiting and negotiating with the two sellers is costless for all parties and whenever indifferent between accepting or rejecting a price offer, each player breaks the indifference in favor of accepting the offer.

- (a) (4 pts) Consider the subgame starting with $\psi_1 = 0$. What is the buyer's willingness to pay for the second parcel given that she failed to purchase the first one?

- (b) (4 pts) Define a strategy profile $(p_2^*(s_2|\psi_1 = 0), p_2^*(b|\psi_1 = 0))$ that constitutes a subgame perfect Nash equilibrium (SPNE) of this subgame. What is the buyer's expected payoff $\pi_b^*(\psi_1 = 0)$ from this subgame?

(c) (4 pts) Consider the subgame starting with $\psi_1 = 1$. What is the buyer's willingness to pay for the second parcel given that she has purchased the first one?

(d) (4 pts) Define a strategy profile $(p_2^*(s_2|\psi_1 = 1), p_2^*(b|\psi_1 = 1))$ that constitutes a SPNE of this subgame.

(e) (4 pts) Consider now the first negotiation and the subgame, in which the buyer is the proposer. Define $p_1^*(b)$ that constitutes a SPNE of this subgame.

(f) (5 pts) Consider now the first negotiation and the subgame, in which the seller is the proposer. What price $p_1^*(s_1)$ constitutes a SPNE in the first negotiation?

(g) (5 pts) Is the SPNE that you found Pareto efficient? Explain. How would your answer change if $v_s > \frac{1}{3}$?

3. (20 pts) Suppose inverse market demand for some good is

$$p = 1 - Q$$

where Q is aggregate output. Firm 1 (the incumbent) is already in this market, and firm 2 (the entrant) must decide whether to enter it. If both firms are in the market, they play a standard simultaneous-move Cournot (quantity-setting) game. If only firm 1 is in the market, then it is a monopoly. Firm 1's cost function is cq_1 where $q_1 \geq 0$ and c is firm 1's marginal cost. The marginal cost is private unverifiable information to firm 1 and can take one of two values- $c \in \{0, 1/2\}$. Firm 2 has beliefs about c summarized by $Pr\{c = 0\} = \alpha$. If firm 2 enters, it incurs a cost of $f = 2/9$ but has production cost of zero. Firm 1 observes firm 2's entry decision, $d_2 \in \{in, out\}$ before the two firms simultaneously choose their output.

- (a) (10 pts) Consider the Bayesian game played between the firms following entry by firm 2. Find a BNE, $q_1^*(in|c), q_2^*$ of the game in terms of α .

(b) (5 pts) Find the SPNE $(q_1^*(d_2|c), d_2, q_2^*)$ of the entire game for 1) $\alpha = 1$ and 2) $\alpha = 0$.

(c) (5 pts) Find the value α^* such that entry occurs in an SPNE only if $\alpha \leq \alpha^*$.

4. **(30 pts)** Consider the following game involving two risk-neutral players. Player 1 may be one of two types. With probability λ she is strong ($\theta = H$) and with probability $1 - \lambda$ she is weak ($\theta = L$). After learning her type, player 1 may bet either one ($s_1 = 1$) or two ($s_1 = 2$) dollars. After observing player 1's bet (but not her type), player 2 decides whether to challenge ($s_2 = c$) or to quit ($s_2 = q$). If player 2 quits, then both players receive payoffs of zero $u_i(s_1, q, \theta) = 0$ for $i = 1, 2$. If player 2 challenges a strong player 1, then player 1 wins the bet and player 2 loses ($u_1(s_1, c, H) = -u_2(s_1, c, H) = s_1$). If player 2 challenges a weak player 1, then player 2 wins the bet and player 1 loses ($u_1(s_1, c, L) = -u_2(s_1, c, L) = -s_1$). Let $\mu(\theta|s_1)$ denote player 2's beliefs about player 1's type after observing s_1 . Also, let $\sigma_1(\theta)$ denote the probability that player 1 takes action $s_1 = 2$ when her type is θ , and let $\sigma_2(s_1)$ denote the probability that player 2 chooses $s_2 = c$ when he observes s_1 .
- (a) (6 pts) Draw the game tree representing this extensive-form game. Be sure and include all terminal nodes and specify all payoffs listing player 1's payoff first and player 2's second as well as clearly label all information sets.

(b) (6 pts) Specify a fully separating PBE $(\sigma_1(\theta), \sigma_2(s_1), \mu(H|s_1))$ in which $\sigma_1(H) = 1$ or explain why no such PBE exists.

(c) (6 pts) Specify a fully separating PNE $(\sigma_1(\theta), \sigma_2(s_1), \mu(H|s_1))$ in which $\sigma_1(H) = 0$ or explain why no such PBE exists.

(d) (6 pts) Suppose that $\lambda < \frac{1}{2}$. Specify a fully pooling PBE $(\sigma_1(\theta), \sigma_2(s_1), \mu(H|s_1))$ in which $\sigma_1(H) = 1$ or explain why no such PBE exists.

(e) (6 pts) Suppose that $\lambda < \frac{1}{2}$. Specify a fully pooling PBE $(\sigma_1(\theta), \sigma_2(s_1), \mu(H|s_1))$ in which $\sigma_1(H) = 0$ or explain why no such PBE exists.